Haoyu Wang

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Education

Beijing Jiaotong University (BJTU), Beijing, China — Sept. 2023 – Jun. 2026

M.S. student in Electronic Information Technology (graduating year), School of Computer Science.

Guilin University of Electronic Technology (GUET), Guilin, China — Sept. 2019 – Jun. 2023

B.Eng. in Robotics Engineering, School of Artificial Intelligence.

Research Interests

Neurosymbolic AI — coupling symbolic / logical priors with deep perception; video event understanding under structured constraints; self-supervised spatiotemporal representations as a substrate for neurosymbolic reasoning.
Generative models — diffusion inversion, disentangled semantic manipulation, training- and tuning-free image editing.
Robust perception — long-tailed, few-shot and dual-view detection, X-ray security inspection.

Publications

Published

- R. Tao, **H. Wang**, Y. Guo, H. Chen, L. Zhang, X. Liu, Y. Wei, Y. Zhao. "Dual-view X-ray Detection: Can AI Detect Prohibited Items from Dual-view X-ray Images like Humans?" *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2025.
- Z. Li, **H. Wang**, W. Wang, C. Tan, Y. Wei, Y. Zhao. "DCI: Dual-Conditional Inversion for Boosting Diffusion-Based Image Editing." *Advances in Neural Information Processing Systems* (**NeurIPS**), 2025.
- Z. Liu, J. Zhang, **H. Wang**, B. Xu, C. Zhang, Y. Zhu, E. Shi. "Scaling Law for Large Wireless Models." *AAAI Conference on Artificial Intelligence* (**AAAI**), 2026.

Under Review

- **H. Wang**, Z. Li, W. Wang, Y. Zhao. "Few-Shot Continual Prohibited Object Detection in X-Ray Security Inspection." Under review at *European Conference on Computer Vision* (**ECCV**), 2026.
- **H. Wang**†, **R. Tao**†, W. Wang, Y. Wei. "PAD-F: Prior-Aware Debiasing Framework for Long-Tailed X-ray Prohibited Item Detection." Under review at *IEEE Transactions on Information Forensics and Security* (**IEEE TIFS**). *(† Equal contribution)*

Projects

Algorithm research on real-world X-Ray prohibited-item detection — TODO.MM – TODO.MM

Prior-aware debiasing for severe head/tail class imbalance; few-shot continual learning for emerging object categories without catastrophic forgetting. Outputs: two first-authored manuscripts above (IEEE TIFS, ECCV 2026).

Honours & Awards

First-Class Academic Scholarship, BJTU *(three consecutive years)*	2023 – 2026
Outstanding Bachelor's Thesis Award (school-level), School of AI, GUET	2023
Outstanding Completion (highest rating), Provincial Undergraduate Innovation Program	TODO

Technical Skills

Python · C/C++ · MATLAB · PyTorch · Linux · Git · LaTeX.